BRAINSTORM GROUP

BRAINSTORM GROUP (BSG)

CHM111
(General Chemistry I)
MOCK TEST 1 [2021-22]

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BRAINSTORM GROUP (BSG) MAT111 TEST 1 (2021/22)

1. In an examination, 18 students passed MAT111, 17 students passed PHY113, 11 students passed both subject. Find the number of students that passed Mat111.
   A. 6 B. 7 C. 8 D. 9

2. If $X = \{2,\ 4,\ 6\}$, determine the power set of X, denoted by $n(P(x))$.
   A. 3 B. 6 C. 8 D. 9

3. Given that $X = \{x : 3 < x < 6\}\ and\ Y = \{x : 4 \leq x \leq 8\}$. $Find\ X \cap Y$.
   A. $\{4\}$ B. $\{4, 5\}$
   C. $\{5, 6\}$ D. $\{4, 5, 6, 7, 8\}$

4. If Q = {all perfect squares less than 30} and P = {all odd numbers from 1 to 10}. Find $(Q \cap P)$.
   A. $\{0, 1, 4, 9\}$ B. $\{1, 4, 9\}$
   C. $\{1, 4\}$ D. $\{4, 9\}$
   E. $\{1, 9\}$

5. If set $F = \{2 < x \leq 5\} \cup \{9 < x < 12\}$. List the members of F.
   A. $\emptyset$
   B. $\{2, 3, 4, 5, 9, 10, 11, 12\}$
   C. $\{3, 4, 5, 6, 9, 10, 11\}$
   D. $\{3, 4, 5, 10, 11\}$

6. Given that $P = \{1,\ 3,\ 5,\ 7\}\ and\ Q = \{2,\ 5,\ 6,\ 8,\ 9\}$. $Find\ (P \cup Q)$.
   A. $\{1, 2, 3, 5, 6, 7, 8, 9\}$
   B. $\{1, 2, 3, 6, 7, 8, 9\}$
   C. $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
   D. $\{3, 5\}$

7. Given that $P = \{1,\ 3,\ 5,\ 7\}\ and\ Q = \{2,\ 5,\ 6,\ 8,\ 9\}$. $Find\ (P \cap Q)$.
   A. $\{\}$ B. $\{3, 5\}$ C. $\{5, 7\}$
   D. $\{3, 5, 7\}$ E. $\{5\}$

8. If $U = \{1,\ 2,\ 3,\ 6,\ 9, 18\}$, $X = \{2,\ 6,\ 18\}$, $Y = \{1,\ 3,\ 6\}$, $list\ the\ elements\ of\ X' \cap Y$.
   A. $\{1, 3\}$ B. $\{1,\ 3,\ 6\}$
   C. $\{1,\ 3,\ 9\}$ D. $\{1,\ 3,\ 6,\ 9\}$

9. In a class of 40 students, 30 take Agriculture and 20 take Physics. If 8 students take neither Agriculture nor Physics, find the number of students who take Agriculture but not Physics.
   A. 2 B. 8 C. 12 D. 18

10. Two sets A and B are such that n(A) = 11 and n(B) = 6. Given that $n(\mathcal{U}) = 15$, find the smallest possible value of $n(A \cap B)$.
    A. 0 B. 17 C. 5 D. 2

11. Two sets A and B are such that $n(A) = 11$ and $n(B) = 6$. Given that $n(\mathcal{U}) = 15$, find the largest possible value of $n(A \cup B)'$
    A. 2 B. 5 C. 6 D. 4

12. All inhabitants of a certain village near the Cameroon border speak either French or English or both. If 64% speak French and 58% speak English. What is percentage speak both languages?
    A. 18 B. 22 C. 23 D. 25

13. The $A \cup B$ is 69, the number of A is 40 and the number of B is 54. How many people are in $A \cap B$?
    A. 15 B. 25 C. 27 D. 29

14. In a class of 50 students, 25 like oranges, 22 like bananas and 4 like both oranges and bananas. How many like neither oranges nor banana?
    A. 1 B. 7 C. 11 D. 12

15. In a class, 16 students offer geography, 18 offer chemistry, 5 offer both and 3 offer neither. How many students are in the class?
    A. 27 B. 33 C. 32 D. 42

16. Given that U = $\{x : x\ is\ an\ integer, 2 \leq x \leq 10\}$, $A = \{x : x\ is\ a\ prime\ number\}$, $B = \{x:\ is\ a\ multiple\ of\ 3\}$. $Find\ (A \cup B)$.
    A. $\{3\}$ B. $\{2, 3, 5\}$
    C. $\{3, 5, 6, 7\}$ D. $\{2, 3, 5, 6, 7, 9\}$

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17. Given that U = {x : x is an integer, $2 \leq x \leq 10$}, $A = \{x : x \text{ is a prime number}\}$, $B = \{x: \text{ is a multiple of } 3\}$. Find $n(A \cup B)$.
A. 1 B. 3 C. 4 D. 6

18. Given that U = {x : x is an integer, $2 \leq x \leq 10$}, $A = \{x : x \text{ is a prime number}\}$, $B = \{x: \text{ is a multiple of } 3\}$.
List the elements of $A' \cap B'$.
A. {2, 3, 5, 6} B. {4, 8, 9, 10}
C. {2, 3, 5, 6, 7, 9} D. {4, 8, 10}

19. In a class of 60 students, 40 speak Hausa while 19 speak Igbo and 7 students speak both languages. How many student speak neither of the languages.
A. 1 B. 8 C. 15 D. 16

20. P and Q are two sets such that is $n(P \cup Q)$ is 50, n(P) is 40 and n(Q) is 32. What is $n(P \cap Q)$?
A. 10 B. 12 C. 18 D. 22

21. Which of the following is not a subset of P if P = {2, 4, 6}.
A. {2, 2, 4} B. {2, 3, 4}
C. {2, 4, 6} D. ∅

22. $(M' \cap N)'$ is equivalent to
A. $(M \cap N')'$ B. $(M' \cap N)$
C. $(M \cup N')$ D. $(M \cup N)$

23. If $P \subset Q, then\ (P \cup Q) \cap P\ is$ is
A. ∅ B. P C. Q D. $\mathcal{U}$

24. The cardinality of the set Z = {M, I, S, S, I, S, S, I, P, P, I} is
A. 4 B. 6 C. 8 D 11

25. $(A \cap A')'$ is equivalent to
A. ∅ B. A C. A' D. $\mathcal{U}$

26. $(A \cup A')$ is equivalent to
A. ∅ B. A C. A' D. $\mathcal{U}$

27. If X = {1, 2, 3, 4} and Y = {3, 5, 6}, the elements of $(X \cap Y) \cup X$ are
A. {3} B. {1, 2, 3, 4}
C. {1, 2, 5, 6} D. {1, 2, 3, 4, 5, 6}

28. If X = {a, e, m, p}, Y = {a, e, i, o, u}, Z = {l, m, o, p, q, r, s, t, u}. The elements of $X \cup (Y \cap Z)$ are
A. {a, e, m, p, o} B. {a, e, m, p, u}
C. {a, e, m, p, o, u, t} D. {a, e, m, p, o, u}

29. If $\mathcal{U}$ = {5, 6, 7, 8}, P = {6, 7} and Q = {6, 8} then $(P \cap Q)'$ is
A. {6} B. {7, 8}
C. {6, 7, 8} D. {5, 7, 8}

30. $P - Q$ is same as
A. $(P \cap Q)$ B. $(P \cap Q)'$
C. $(P \cap Q')$ D. $(P' \cap Q)$

31. Given $n(A - B) = 18, n(A \cup B) = 70\ and\ n(A \cap B) = 25$. Find $n(B)$.
A. 27 B. 43 C. 45 D. 52

32. In a class of 500 students, 270 offer Chemistry, 250 offer Geography, 110 offer neither. Find the number of students who offer Chemistry and Geography.
A. 110 B. 120
C. 130 D. 140

33. Sets A and B are defined by A = {3, 5, 7, 8} and B = {x, y, z}. Is $x \in A$?
A. True B. False

34. In a Survey of 23 young man, it was recorded that 6 do play Football, 8 play Basketball and 12 play none. Find the number of young men who play only Basketball.
A. 3 B. 5 C. 7 D. 11

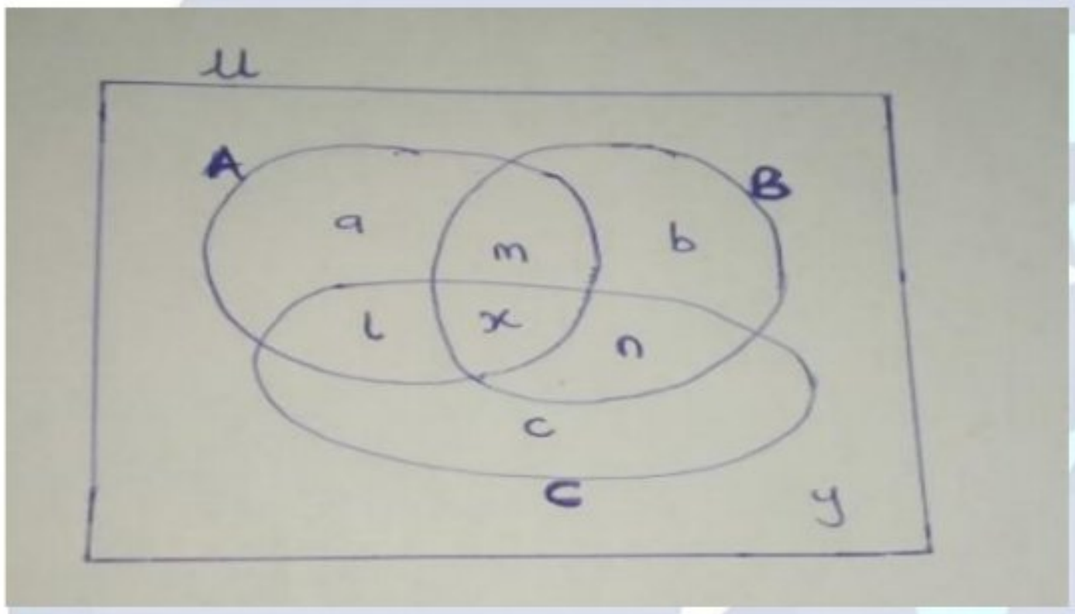
35. In a class of 40 students, 32 offer Mathematics, 24 offer Biology and 4 offer neither Biology nor Mathematics. Find the number of students who offer only one subject.
A. 4 B. 12 C. 16 D. 20

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36. Given the set $\{R = x : x \in \mathbb{N}, x < 5\}$. List elements of R.
    A. {... -1, 0, 1, 2, 3, 4, 5}
    B. {0, 1, 2, 3, 4, 5}
    C. {... -1, 0, 1, 2, 3, 4}
    D. {1, 2, 3, 4}

37. In a class of 50 students, 31 offer Chemistry, 27 offer Biology while 3 neither. Find the number of students that offer chemistry only.
    A. 11 B. 14 C. 16 D. 20



The Venn diagram above represents a group of students offering course A, B and C. Use the diagram above to answer question 38 – 40

38. The portion y can be represented as
    A. $(A \cup B \cup C)$
    B. $(A \cap B \cap C)$
    C. $(A \cup B \cup C)'$
    D. $(A \cap B \cap C)'$

39. Which of the following corresponds to number of students that offer only two courses?
    A. (a + b + c)
    B. (l + m + n)
    C. (x)
    D. (y)

40. The number of students that offer at least two courses is
    A. (a + b + c + l + m + n)
    B. (l + m + n + x)
    C. (a + b + c + x)
    D. (x – y)

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ANSWERS							
1.	B	11.	D	21.	B	31.	D
2.	C	12.	B	22.	C	32.	C
3.	B	13.	B	23.	B	33.	B
4.	E	14.	B	24.	A	34.	B
5.	D	15.	C	25.	D	35.	C
6.	A	16.	D	26.	D	36.	D
7.	E	17.	D	27.	B	37.	D
8.	A	18.	D	28.	D	38.	C
9.	C	19.	B	29.	D	39.	B
10.	D	20.	D	30.	C	40.	B

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